CS24307 • COMPLETE MASTER GUIDE

Artificial Intelligence (CS24307) 10-Page Master Revision

High-Yield Formulas, Search Matrices, Logic Rules, Bayes Formulas & Flashcards

Page 1: AI Master Mental Model & Environmental Taxonomy

Core Paradigm

Rational Agent: Selects action $a \in \mathcal{A}$ maximizing expected performance measure P given historical percept sequence $\mathcal{P}^*$.

Environment	Observability	Determinism	Episodic/Seq	Static/Dyn	Discrete/Cont	Agents
Chess	Fully	Deterministic	Sequential	Static	Discrete	Multi (Adversarial)
Poker	Partially	Stochastic	Sequential	Static	Discrete	Multi (Adversarial)
Self-Driving	Partially	Stochastic	Sequential	Dynamic	Continuous	Multi
Medical Diagn.	Partially	Stochastic	Sequential	Dynamic	Continuous	Single

Page 2: Uninformed vs. Informed Search Complexity Matrix

Algorithm	Data Structure	Time Complexity	Space Complexity	Complete?	Optimal?
BFS	FIFO Queue	$O(b^d)$	$O(b^d)$	Yes	Yes (uniform costs)
DFS	LIFO Stack	$O(b^m)$	$O(b \cdot m)$	No (infinite trees)	No
UCS	Priority Queue	$O(b^{1+\lfloor C^*/\epsilon \rfloor})$	$O(b^{1+\lfloor C^*/\epsilon \rfloor})$	Yes	**Yes (Cost optimal)**
IDDFS	LIFO Stack	$O(b^d)$	$\mathbf{O(b \cdot d)}$	Yes	Yes (uniform costs)
A^* **Search**	Priority Queue	$O(b^d)$	$O(b^d)$	Yes	**Yes (if h admissible)**

Page 3: A^* Search Admissibility & Consistency Theorems

$$\text{Admissibility: } 0 \le h(n) \le h^*(n) \implies A^* \text{ Tree Search is Optimal}$$

$$\text{Consistency: } h(n) \le c(n, a, n') + h(n') \implies A^* \text{ Graph Search is Optimal (No node re-expansion!)}$$

Page 4: Adversarial Minimax & Alpha-Beta Pruning Rules

$$\alpha \leftarrow \max(\alpha, \text{child\_val}) \quad (\text{Initialized to } -\infty)$$

$$\beta \leftarrow \min(\beta, \text{child\_val}) \quad (\text{Initialized to } +\infty)$$

$$\text{PRUNE SUBTREE IF } \alpha \geq \beta$$

Page 5: Propositional & First-Order Logic (FOL) Transformation Rules

$$\alpha \rightarrow \beta \equiv \neg\alpha \vee \beta \qquad \alpha \leftrightarrow \beta \equiv (\neg\alpha \vee \beta) \wedge (\neg\beta \vee \alpha)$$

$$\neg(\forall x P(x)) \equiv \exists x \neg P(x) \qquad \neg(\exists x P(x)) \equiv \forall x \neg P(x)$$

$$\text{Resolution: } \frac{A \vee B, \quad \neg B \vee C}{A \vee C}$$

Page 6: Classical Planning STRIPS & Graphplan Mutexes

3 Mutex Conditions

Inconsistent Effects (Add vs Delete), Interference (Delete vs Precond), Competing Needs (Mutex Preconditions).

Page 7: Probability, Bayes' Rule & Bayesian Networks

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)} \qquad P(X_1, \ldots, X_n) = \prod_{i=1}^{n} P(X_i \mid \text{Parents}(X_i))$$

Page 8: Decision Tree Induction (ID3 Entropy & Gain)

$$H(S) = -\sum p_i \log_2(p_i) \qquad \text{Gain}(S, A) = H(S) - \sum \frac{|S_v|}{|S|} H(S_v)$$

Page 9: Multi-Layer Perceptrons & Backpropagation Gradient

$$\delta_k = (y_k - \hat{y}_k)\sigma'(z_k) \qquad \delta_j = \sigma'(z_j)\sum_k \delta_k w_{jk} \qquad \Delta w = \eta\delta a_{\text{in}}$$

Page 10: AI Final High-Yield Exam Checklist & Formulas

10 Instant Revision Points

1. A^* with $h = 0$ becomes Uniform Cost Search.
2. A^* with $g = 0$ becomes Greedy Best-First Search.
3. Iterative Deepening DFS uses $O(bd)$ memory and visits nodes at depth d only once.
4. Alpha-Beta does not affect final minimax value at root.
5. Skolemization replaces $\exists x$ with constant or Skolem function $f(y_1 \dots y_k)$.
6. Unification requires variables to be standardized apart.
7. d-separation isolates nodes given evidence variables.
8. Backpropagation computes exact gradient via dynamic programming chain rule.
9. ReLU eliminates vanishing gradient for positive activations.
10. Softmax produces valid categorical probability distribution.